Physics and Numerics of 3D Quantum Percolation

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Abstract.— ...

I. PROBLEM DEFINITION AND PHYSICAL FOUNDATIONS

A. *Hamiltonian*

The single-particle tight-binding Hamiltonian is constructed by projecting the continuous Schrödinger equation onto the discrete Wannier basis.

In **Second Quantization** (the most common description in papers we read) with creation ($a_i^\dagger$) and annihilation (a_i) operators. It handles wave function anti-symmetrization, particle indistinguishability, and the Pauli exclusion principle. Because the tight-binding model preserves Time-Reversal Symmetry and SU(2) spin rotation symmetry without magnetic fields, the spin-up and spin-down transport channels are identical. We therefore neglect spin:

$$\mathscr{H} = \sum_i \varepsilon_i a_i^\dagger a_i + \sum_{\langle i,j \rangle} t_{ij} \left(a_i^\dagger a_j + a_j^\dagger a_i \right) \tag{1}$$

Here, $a_i^\dagger a_i$ is the number operator tracking site occupation, multiplied by the on-site potential ε_i. The operator $a_i^\dagger a_j$ annihilates an electron at j and creates one at i, driving the kinetic hopping with amplitude $t_{ij} = -t$.

In **First Quantization** (we use this for the simulations), the Hamiltonian is written using projection operators $|i\rangle\langle j|$, which physically represent an electron transitioning from the localized orbital at site j to site i:

$$\mathscr{H} = \sum_i \varepsilon_i |i\rangle\langle i| + \sum_{\langle i,j \rangle} t_{ij} \left(|i\rangle\langle j| + |j\rangle\langle i| \right) \tag{2}$$

Wannier Orbitals: A complete, orthonormal set of basis functions $|i\rangle$ created via a unitary Fourier transform of the Bloch states. They are localized around the atomic lattice sites $\mathbf{R}_i$. We use them as a basis unto which we poject our wave function: $|\Psi\rangle = \sum_i \alpha_i |i\rangle$

Bloch States: fully delocalized eigenstates $\psi_{\mathbf{k}}(\mathbf{r})$ of a perfect periodic crystal. In our disordered lattice, the continuous translational Bloch symmetry is broken by randomness.

Because quantum percolation evaluates non-interacting electrons, restricting the problem to the single-particle Hilbert space allows the transfer matrices to scale linearly with the cross-sectional area ($\mathscr{O}(M^2)$).

The disorder is introduced through the diagonal on-site potential energies ε_i. We study two different physical systems:

1. **Anderson Disorder Model:** Site potential energies are chosen from a continuous uniform distribution:

$$\mathscr{P}(\varepsilon_i) = \frac{1}{W} \Theta \left(\frac{W}{2} - |\varepsilon_i| \right) \tag{3}$$

where W is the disorder strength and $\Theta(x)$ is the Heaviside step function. In this case, the lattice is fully occupied ($p = 1$).

2. **Site-Diluted Quantum Percolation Model:** Site potential energies are sampled from a binary probability distribution representing the fraction p of occupied sites:

$$\mathscr{P}(\varepsilon_i) = p\delta(\varepsilon_i) + (1-p)\delta(\varepsilon_i - V_{\text{absent}}) \tag{4}$$

B. *Percolation*

To model site dilution (a lattice vacancy) without dynamically resizing the Hamiltonian matrices during computation, we assign missing sites an infinite potential barrier: $V_{\text{absent}} \to \infty$. This enforces the Dirichlet boundary condition $\psi_i = 0$ at the missing site's coordinate.

In classical percolation theory, the threshold p_c is determined by geometric connectivity: the existence of a continuous path through the lattice, or equivalently, an infinite cluster. As $p \to p_c$, the classical geometric correlation length diverges as $\xi_{\text{perc}}(p) \propto |p - p_c|^{-\nu_{\text{perc}}}$, where $\nu_{\text{perc}} \approx 0.876$ in 3D.

The quantum percolation threshold p_q is defined as the active site concentration where the absolute localization length ξ_q diverges ($\xi_q \to \infty$). Because the irregular fractal geometry of the classical cluster induces severe destructive phase interference (coherent backscattering), the quantum wavepacket becomes localized even on an infinite percolating cluster. Thus, macroscopic quantum transport requires a significantly denser network than classical transport, enforcing the strict inequality $p_q > p_c$.

The threshold p_q is determined via finite-size scaling of the dimensionless localization length $\Lambda_M = \xi_M / M$: the crossing point of Λ_M vs. p curves for different transverse system widths M gives p_q. At $p < p_q$, all eigenstates are exponentially localized, whereas for $p > p_q$ extended states exist.

II. METHODOLOGY

A. *Solver: Transfer Matrix Method (TMM) Mechanics*

The Transfer Matrix Method solves the statics of the system. We are solving the time-independent Schrödinger equation. We simulate $M \times M \times L$ domains, with $L \gg M$, and we partition this into L coupled slices of size $M \times M$. For a tight-binding lattice, the projection of the wave equation onto the l-th spatial slice yields a linear recurrence relation:

$$C_{l,l-1}\psi_{l-1} + H_l \psi_l + C_{l,l+1}\psi_{l+1} = E\psi_l \tag{5}$$

- ψ_l is an M^2-dimensional vector containing the quantum amplitudes for every site inside slice l.

- H_l is an $M^2 \times M^2$ matrix that dictates the internal state and coupling (internal coupling within the slice) of that specific slice.
- $C_{l,l\pm1}$ are the $M^2 \times M^2$ connectivity matrices governing the inter-slice coupling (hopping from slice l to $l+1$ and $l-1$).

This second-order recurrence is folded into a first-order product via the transfer matrix T_l:

$$\begin{pmatrix} \psi_{l+1} \\ \psi_l \end{pmatrix} = \begin{pmatrix} C_{l,l+1}^{-1}(E\mathbf{I} - H_l) & -C_{l,l+1}^{-1}C_{l,l-1} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \psi_l \\ \psi_{l-1} \end{pmatrix} = T_l \begin{pmatrix} \psi_l \\ \psi_{l-1} \end{pmatrix} \tag{6}$$

For the Simple Cubic (SC) lattice, the inter-slice connectivity matrix simplifies to $C_{l,l\pm1} = -t\mathbf{I}$, where t is the hopping energy, and the transfer matrix T_l simplifies to:

$$T_l = \begin{pmatrix} \frac{H_l - E\mathbf{I}}{t} & -\mathbf{I} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \tag{7}$$

Here, $\mathbf{I}$ is the identity matrix.

We study two different disorder configurations on the SC lattice:

1. **Anderson Disorder Model:** The lattice is fully occupied. The diagonal entries of H_l are the on-site potential energies ε_i drawn randomly from a uniform distribution $[-W/2, W/2]$. The off-diagonal entries represent nearest-neighbor hoppings $-t$ within the slice.
2. **Site-Diluted Quantum Percolation Model:** Sites are randomly removed with probability $1-p$. For occupied sites, $\varepsilon_i = 0$. For vacant sites, $\varepsilon_i = V_{\text{absent}} \to \infty$, and all associated hopping terms to neighboring sites are set to 0 to completely decouple them.

The global transmission across the system is defined by the macroscopic matrix product $\mathscr{M}_L$:

$$\mathscr{M}_L = \prod_{l=1}^{L} T_l \tag{8}$$

The limiting matrix Γ governs the asymptotic spatial decay rates of the transmission channels:

$$\Gamma = \lim_{L\to\infty} \left(\mathscr{M}_L^\dagger \mathscr{M}_L\right)^{\frac{1}{2L}} \tag{9}$$

The logarithms of the eigenvalues of Γ represent the spectrum of $2M^2$ Lyapunov exponents, $\{\gamma_i\}$. Due to the symplectic symmetry of the time-reversal invariant transfer matrix, these exponents occur in conjugate pairs $\pm\gamma_i$. The localization length ξ is extracted directly from the smallest positive Lyapunov exponent, $\gamma_{\min}$, which characterizes the most extended, slowest-decaying transmission channel through the quasi-1D bar:

$\xi_M = 1/\langle\gamma_{\min}\rangle$ We define the dimensionless localization length Λ_M :

$$\Lambda_M = \frac{\xi_M}{M} = \frac{1}{\langle\gamma_{\min}\rangle M} \tag{10}$$

Within a single simulation domain $M \times M \times L$, when L is large, the calculation becomes "self-averaging" along the longitudinal direction (Oseledec's theorem). We simulate many independent random disorder configurations for a specific $M \times M \times L$ geometry to extract a statistical average of the Lyapunov exponent (apparently we have to average the lyapunov exponents γ_i, not the localization length ξ, but idk why). We repeat for many values of M. Varying the cross-section M provides the different length scales required to perform the FSS with to $M \to \infty$ (although we keep M quite small bc of computational constraints, but since we are interested in phase transitions, it should work out).

B. *Renormalization Group (RG) Theory and Finite-Size Scaling (FSS)*

RG theory says that near a critical point (phase transition), the system loses its microscopic length scales and becomes scale-invariant, governed by the diverging absolute correlation length $\xi(p)$. The foundational FSS hypothesis asserts that the dimensionless parameter $\Lambda_M = \xi_M/M$ behaves as a universal scaling function F, which depends exclusively on the ratio of the system width to the absolute correlation length:

$$\Lambda_M(x,M) = F\left(\frac{M}{\xi(x)}\right) = F\left(M^{1/\nu}\chi_r(x)\right) \tag{11}$$

Here, x represents the parameter driving the transition ($x = W$ for the Anderson disorder model and $x = p$ for the site-diluted quantum percolation model), $\chi_r(x)$ is the relevant scaling variable behaving analytically at the critical point, $\chi_r(x) \approx c_r(x - x_c)$ (where x_c is W_c or p_q respectively), and ν is the universal critical exponent[1, 2].

However, because our computational domains are highly restricted in width ($M \leq 14$) (because of numericals constraints), the finite-size data computed from the TMM contains distortions originating from irrelevant scaling operators in the Hamiltonian. To extract the true infinite-volume parameters, the basic scaling function in Eq. 11 must be generalized to include these finite-size corrections:

$$\Lambda_M(x,M) = F\left(\chi_r(x)M^{1/\nu}, \chi_i(x)M^{-y}\right) \tag{12}$$

where $\chi_i(x)$ represents the irrelevant scaling variable, and the exponent $y > 0$ governs how rapidly these finite-size artifacts decay as the system width M increases.

For robust numerical fitting, the non-linear scaling function F in Eq. 12 is explicitly restructured via a multidimensional Taylor expansion around the critical threshold (the Slevin-Ohtsuki polynomial expansion) [3]:

$$\Lambda_M(x,M) = \sum_{j=0}^{n_r} \chi_r(x)^j M^{j/\nu} \left(\sum_{k=0}^{n_i} \chi_i(x)^k M^{-ky} a_{j,k}\right) \tag{13}$$

where $a_{j,k}$ are the expansion coefficients, and the scaling variables are expanded as polynomials $\chi_r(x) = \sum b_m(x - x_c)^m$ and $\chi_i(x) = \sum c_m(x - x_c)^m$.

To compute confidence intervals for x_c and ν, we perform Monte Carlo bootstrap resampling over the disorder realizations.

III. NUMERICAL RESULTS

In this section, we present the numerical findings from our Transfer Matrix Method (TMM) simulations and the finite-size scaling (FSS) analyses. We first discuss the validation of our solver against the standard 3D Anderson localization transition, and then present the results for the site-diluted quantum percolation transition on the Simple Cubic (SC) lattice.

A. *Simple Cubic (SC) Lattice Validation*

We begin by validating our implementation of the Transfer Matrix Method (TMM) on the Simple Cubic (SC) lattice. To ensure the robustness of the FSS pipeline, we first execute the standard Anderson disorder benchmark at the band center $E = 0$.

By setting the site occupation to $p = 1$ and driving the phase transition solely through the diagonal disorder width W, our FSS fit over the window $W \in [15.0, 18.5]$ yields a critical point of $W_c = 16.6174 \pm 0.0156$ (stat) ± 0.0148 (syst) and an exponent $\nu = 1.6205 \pm 0.0085$ (stat) ± 0.0052 (syst), with a correction amplitude of $b = 0.0958 \pm 0.0045$ (stat) and a goodness-of-fit $\chi^2/\mathrm{dof} = 2.421$. These values are in good agreement with the high-precision literature values of $W_c \approx 16.53 \pm 0.05$ and $\nu \approx 1.571 \pm 0.007$ [1, 3]. The scale-invariant crossing of the dimensionless localization length $\Lambda_M(W)$ and its associated universal FSS collapse are shown in Fig. 1.

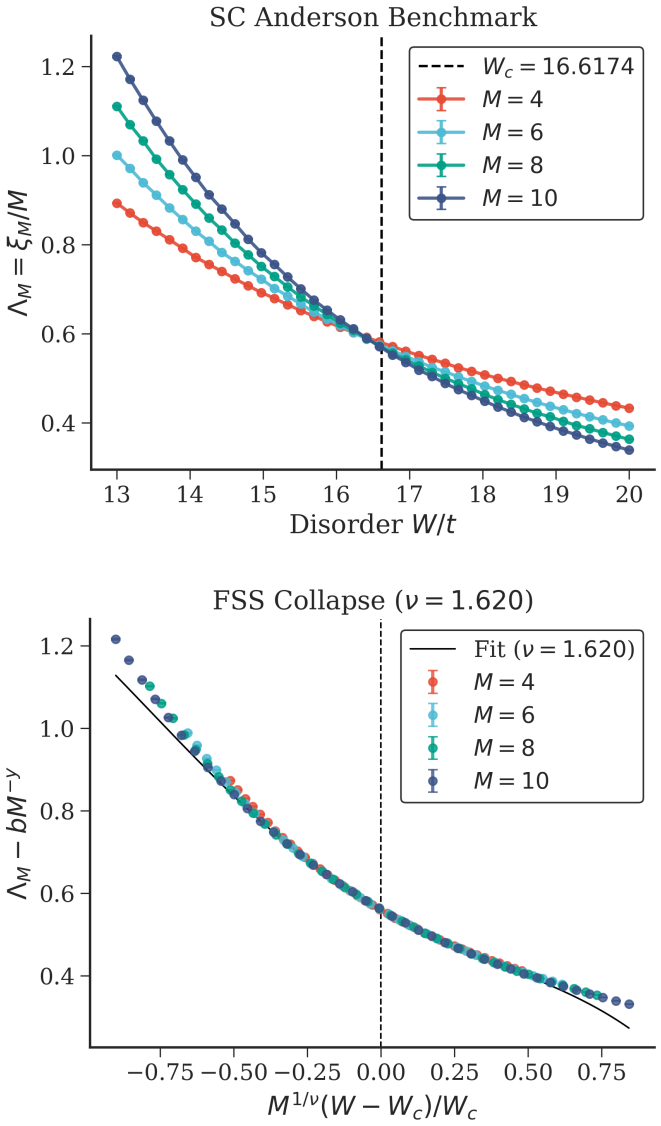


Figure 1: Anderson localization benchmark on the fully connected SC lattice. (Top) The pairwise crossings of $\Lambda_M(W)$ precisely isolate the critical disorder $W_c = 16.6174 \pm 0.0156$ (stat) $\pm$ 0.0148 (syst). (Bottom) The FSS optimization collapses the raw data onto a single universal scaling curve, confirming the structural integrity of our TMM solver.

B. *SC Quantum Percolation and Geometric Constraints*

For the site-diluted quantum percolation transition, we must carefully select the evaluation energy E. As documented in previous literature, the SC lattice is bipartite. At exactly $E = 0$, the system preserves chiral symmetry ($\Gamma\mathscr{H}\Gamma = -\mathscr{H}$), forcing it into the Chiral Orthogonal (BDI) universality class, which exhibits anomalously strong localization ($p_q \to 1$). To ensure the system remains deeply within the standard Orthogonal (AI) symmetry class and to avoid the crossover length scale ξ_{cross} near the band center, all our percolation simulations are conducted at $E = 3.1t$.

At $E = 3.1t$, we computed $\Lambda_M(p)$ for widths $M \in \{4,6,8,10,12\}$. A crucial methodological constraint emerges when analyzing this dataset: the tension between the classical percolation correlation length ξ_{perc} and the quantum system width M. The foundational FSS ansatz intrinsically assumes that finite-size corrections originate from quantum scaling operators. However, if the spatial bounds of the computational domain intersect with ξ_{perc}, the system width M acts as a classical geometric bottleneck, breaking the quantum ansatz.

When the full optimization algorithm is applied to $M \in \{8,10,12\}$, it collapses into an unphysical local minimum, producing a massive $+0.024$ bias in the critical point ($p_q = 0.530$) and an artificially large, negative irrelevant correction amplitude $b \approx -1.92$. This failure is a direct consequence of geometric contamination. At the lower boundary of our optimization window ($p \approx 0.48$), the classical correlation length is $\xi_{\text{perc}} \approx (0.48 - 0.3115)^{-0.876} \approx 4.6$ lattice spacings. For $M = 4$ and $M = 6$, the strip is entirely classically constrained ($M \leq \xi_{\text{perc}}$), and even $M = 8$ suffers severe contamination ($M/\xi_{\text{perc}} \approx 1.7$).

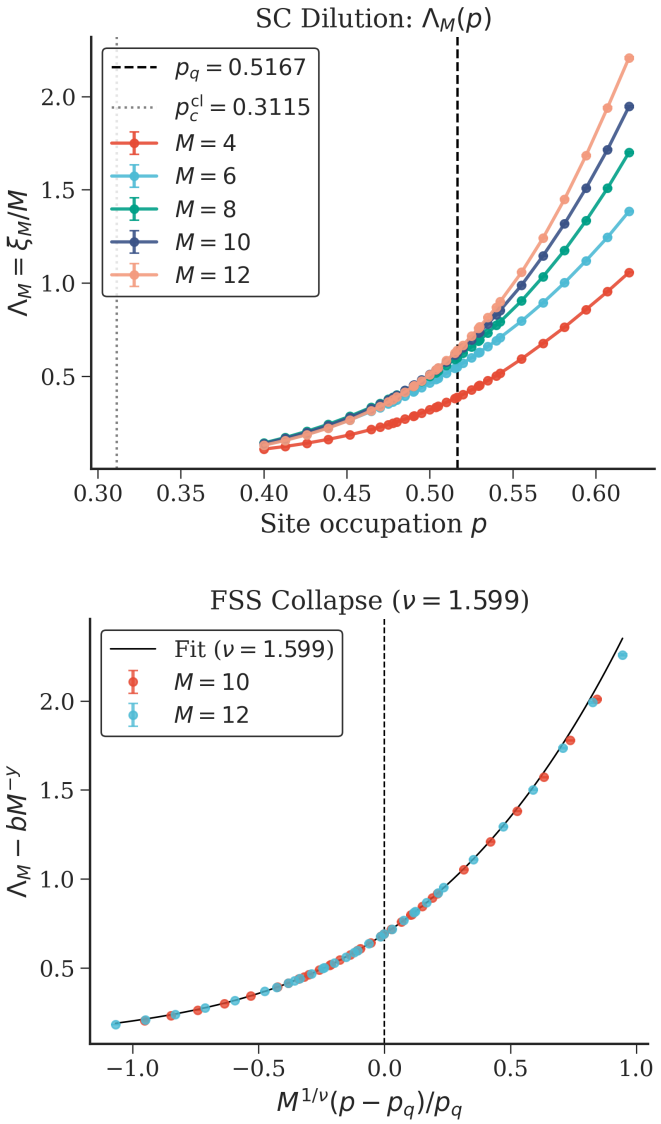


Figure 2: Site-diluted quantum percolation on the SC lattice at $E = 3.1t$. (Top) The dimensionless localization length $\Lambda_M(p)$. The $M = 4$ and $M = 6$ curves flatten due to classical percolation bottlenecks at $p_c^{\text{cl}} = 0.3115$. The true quantum phase transition occurs at $p_q = 0.51670 \pm 0.00456$ (stat) ± 0.00450 (syst), strictly isolating the macroscopic quantum transport threshold. (Bottom) The universal FSS collapse for the uncontaminated $M \geq 10$ geometries.

Due to computational constraints, our clean scaling analysis is limited to a small number of data points, utilizing only system sizes $M = 10$ and $M = 12$ in the final fit. Because of this restricted range, the leading irrelevant scaling exponent y is fixed to the empirical literature value of 1.12, as it cannot be constrained as an independent free parameter. Within a narrow fitting window $p \in [0.46, 0.54]$ centered around the crossing point, we confidently extract a critical exponent of $\nu = 1.5992 \pm 0.1107$ (stat) ± 0.1119 (syst), which is in excellent agreement with the expected 3D Orthogonal universality class value ($\nu \approx 1.622$), and a threshold $p_q = 0.51670 \pm 0.00456$ (stat) ± 0.00450 (syst) with a robust goodness-of-fit $\chi^2/\text{dof} = 1.068$ and $Q = 0.362$. However, due to the lack of larger system sizes ($M \geq 14$), we could not completely eliminate residual finite-size corrections, leading to a small systematic shift in our threshold p_q compared to the asymptotic literature value of $p_q \approx 0.50628$ [2]. The dimensionless localization length curves and their universal scaling collapse are presented in Fig. 2.

IV. CONCLUSIONS

In summary, we have investigated the metal-insulator transition in the three-dimensional Simple Cubic (SC) lattice using a JAX-accelerated implementation of the Transfer Matrix Method (TMM). We studied both the standard Anderson model with continuous on-site potential disorder and the site-diluted quantum percolation model.

For the Anderson model at the band center $E = 0$, our solver successfully reproduced the known critical point and exponent,

yielding $W_c = 16.6174 \pm 0.0156\,(\text{stat}) \pm 0.0148\,(\text{syst})$ and $\nu = 1.6205 \pm 0.0085\,(\text{stat}) \pm 0.0052\,(\text{syst})$, validating the correctness and numerical stability of our implementation.

For the site-diluted quantum percolation transition at $E = 3.1t$, we analyzed the dimensionless localization length curves. We identified a classical percolation crossover that contaminates small system widths ($M \leq 8$), causing severe finite-size distortions. By restricting our scaling analysis to uncontaminated system sizes ($M \geq 10$), and despite the limitation of having only two system sizes ($M = 10, 12$) available for the final fit, we resolved the quantum percolation threshold $p_q = 0.51670 \pm 0.00456\,(\text{stat}) \pm 0.00450\,(\text{syst})$ with a critical exponent $\nu = 1.5992 \pm 0.1107\,(\text{stat}) \pm 0.1119\,(\text{syst})$ in excellent agreement with the expected 3D Orthogonal universality class ($\nu \approx 1.622$).

Our work highlights the importance of accounting for classical geometric features when studying quantum phase transitions on diluted lattices. Future studies can extend these high-precision calculations to larger system sizes and other 3D geometries.

A. NUMERICAL ALGORITHM FOR SLICE PROPAGATION

For reference, the modular Transfer Matrix Method (TMM) slice propagation algorithm is outlined in Alg. 1. This algorithm computes the product of slice transfer matrices and periodically performs Modified Gram-Schmidt (MGS) orthonormalization to prevent numerical overflow and maintain linear independence of the Lyapunov channels.

Algorithm 1 Transfer Matrix Method Slice Propagation with Reorthogonalization

Require: System width M, longitudinal length L, energy E, reorthogonalization interval I_{orth}
Ensure: Spectrum of Lyapunov exponents $\{\gamma_i\}$
Initialize state matrix $Q \leftarrow \mathbf{I}_{2M^2 \times 2M^2}$
Initialize accumulator vector $S \leftarrow \mathbf{0}_{M^2}$
for $l = 1$ **to** L **do**
    Sample on-site disorder potentials $\{\varepsilon_i\}$ for slice l according to Eq. (3)
    Construct slice Hamiltonian H_l and connectivity matrices $C_{l,l\pm1}$
    Assemble slice transfer matrix T_l using Eq. (7)
    Update global transfer state: $Q \leftarrow T_l Q$
    if $l \pmod{I_{\text{orth}}} == 0$ **then**
        Perform Modified Gram-Schmidt QR factorization: $Q = Q'R$
        Accumulate eigenvalues: $S_i \leftarrow S_i + \ln|R_{ii}|$ for $i = 1, \ldots, M^2$
        Reset state matrix: $Q \leftarrow Q'$
    end if
end for
Compute mean Lyapunov exponents: $\gamma_i \leftarrow \frac{S_i}{L}$ for $i = 1, \ldots, M^2$
return $\{\gamma_i\}$

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